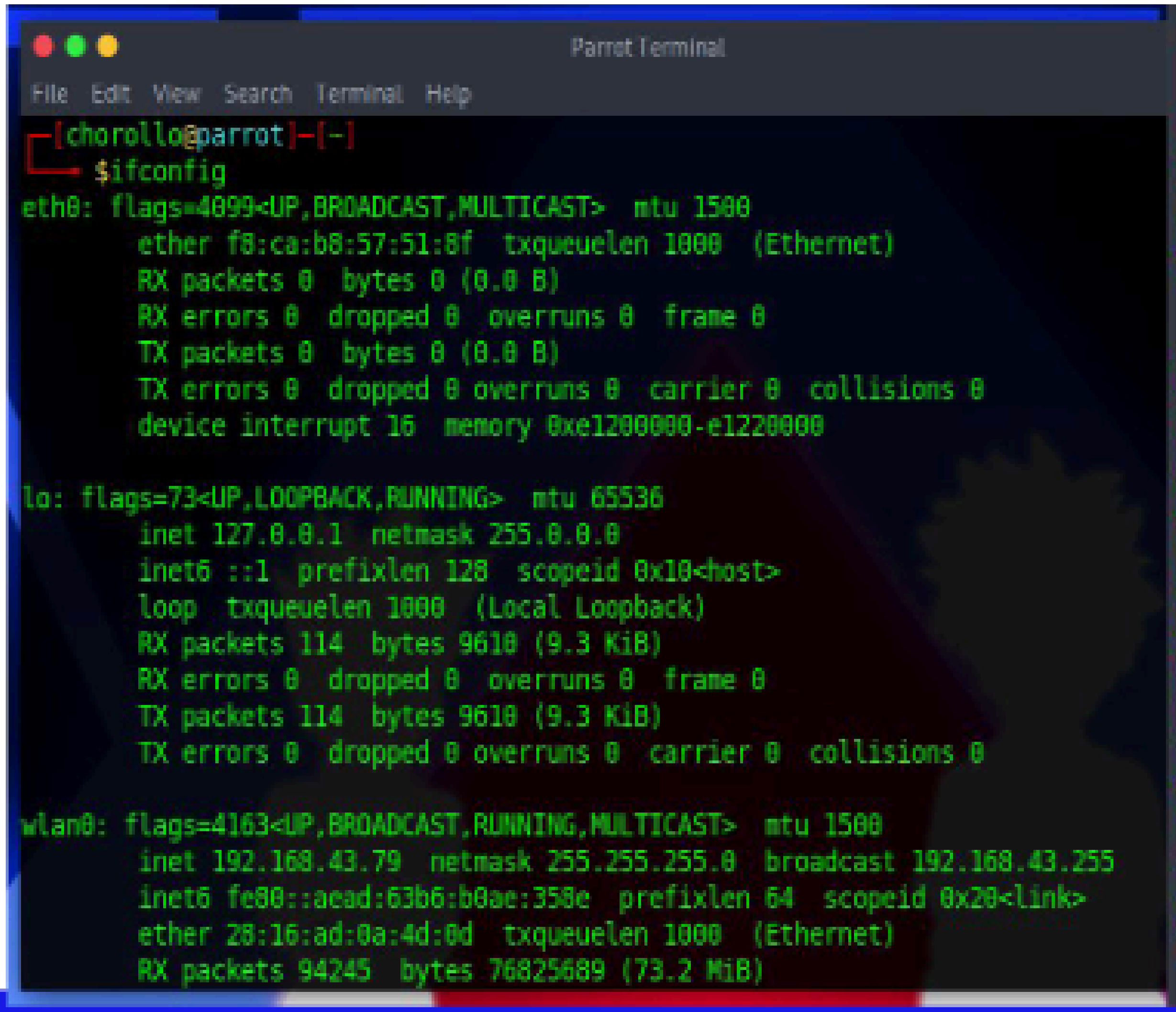
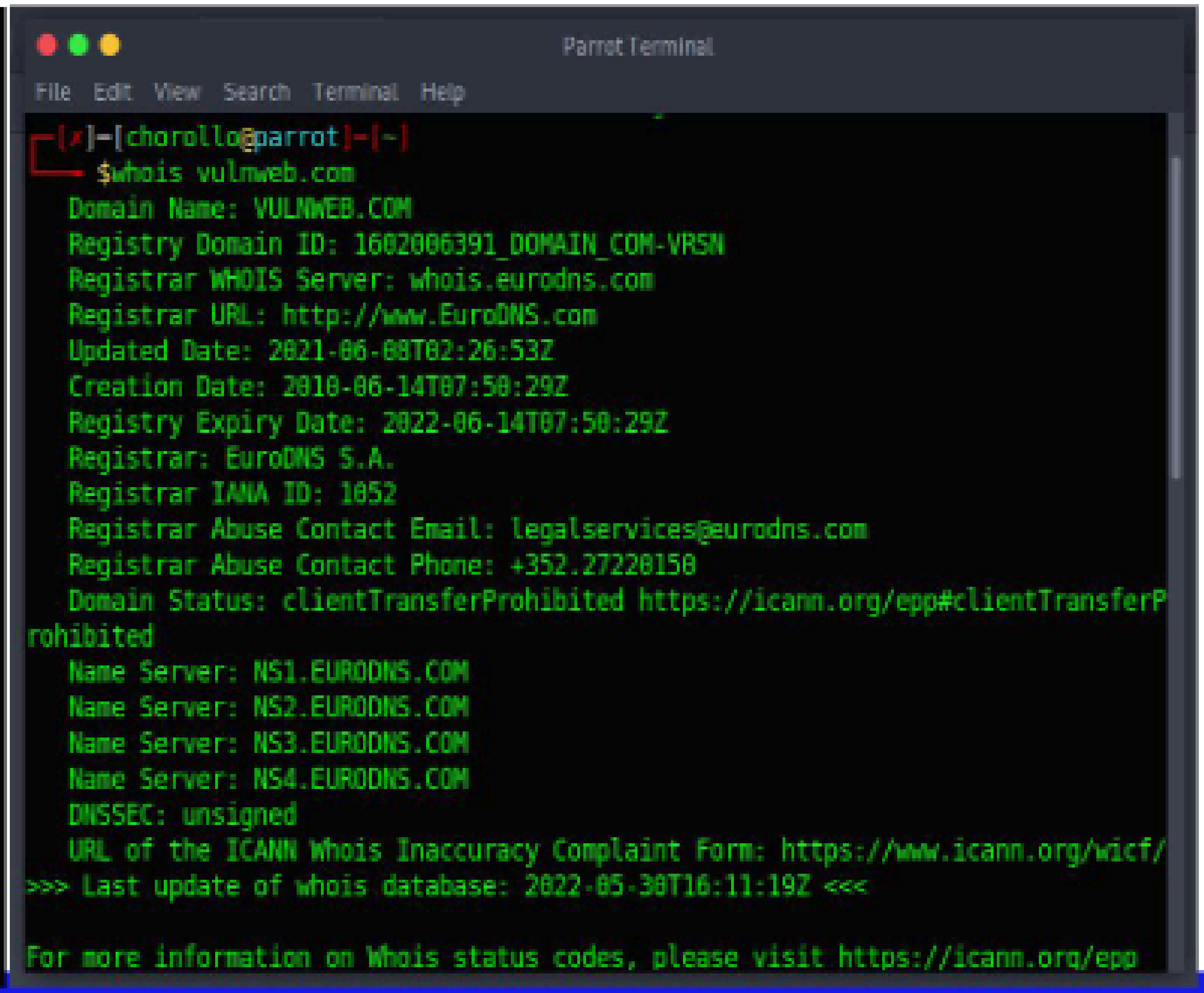
Module 3: Introduction to Network Security

Topic 2: Network Security Tools

Important Networking Tools and Commands

- Ifconfig
- Ping
- Nslookup
- Netstat
- Whois





Important Networking Tools and Commands

- **Nmap:**
  - Nmap is a powerful network scanning and host discovery, OS fingerprinting tool.

Nmap usage:
  - Basic Scan: nmap 192.168.1.5

```
┌─[chorollo@parrot]─[~]
└──╼ $nmap scanme.nmap.org
Starting Nmap 7.80 ( https://nmap.org ) at 2022-05-31 17:05 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.32s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2
f
Not shown: 996 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
9929/tcp  open  nping-echo
31337/tcp open  Elite
```

Network Security Devices

Firewall Products:

- **Fortinet FortiGate.**
- **Palo Alto Networks PA-7000 Series.**
- **Cisco ASA 5500-X.**

Network Security Protections

- **Access control:**
  - The first step in network security implementations is to conduct a powerful access control mechanism for preventing unauthorized parts from accessing network resources and to manage the access level for employees in a suitable way.

- **Firewalls:**
  - Firewalls are a very common protection mechanism used in network security implementations,and firewall can be a software or a hardware, Firewalls are monitoring protections that monitor the income or outcome traffic, it can also filter data and isolate malicious traffic.
- **Intrusion prevention systems:**
  - Also known as IPS, are network security technologies that similarly to firewalls also monitors traffic and recognize exploits and prevent them.

Network Security Protections

Firewalls Types:

- **Packet Filtering Firewall:**
  - It is a fairly old firewall and it operates and filters packets based on the source and destination addresses and ports, it also lacks logging capabilities which makes it impractical in many situations.
- **Application Level Firewall:**
  - Also can be referred to as Proxy, this type of firewalls doesn’t filter packets based on the ports or addresses but it has also the ability to work with the actual content of the packets such as filtering HTTP headers.
- **Next Generation Firewalls:**
  - is a part of the third generation of firewall technology, combining a traditional firewall with other network device filtering functions, such as an application firewall using in-line deep packet inspection, an intrusion prevention system.

Network Security Protections

Type of Intrusion Prevention Systems:

- **Snort:**
  - Snort is an open-source intrusion detection and prevention system (IDS & IPS) and it uses rules to detect malicious activity such as anomaly, protocol and signature detection.
- **Snort core features**:
  - Real time traffic monitoring.
  - Packet logger mode.
  - OS fingerprinting.
  - Flexible rule language.
  - Protocol analysis.
- **Snort modes:**
  - **Sniffer mode:** collect the IP packets and displays them on the console.
  - **Packet logger mode:** log all the IP packets the goes through the network.
  - **Intrusion detection system mode:** uses the built in rules to filter malicious packets.

Network Security Protections

- **Suricate:**
  - It is an open-source threat detection engine that combines intrusion detection and prevention technologies beside network security monitoring.
- **Suricata core features:**
  - Powerful integration with different commercial and free security solutions.
  - High and stable performance.
  - Automatic protocol detection.
  - Off-line analysis of PCAP files.
  - Ability to use LUA (programming language) scri

Network Monitoring and Analysis

GUI Tools	Command Line Tools
WireShark	Tshark
Networkminer	TCPDUMP

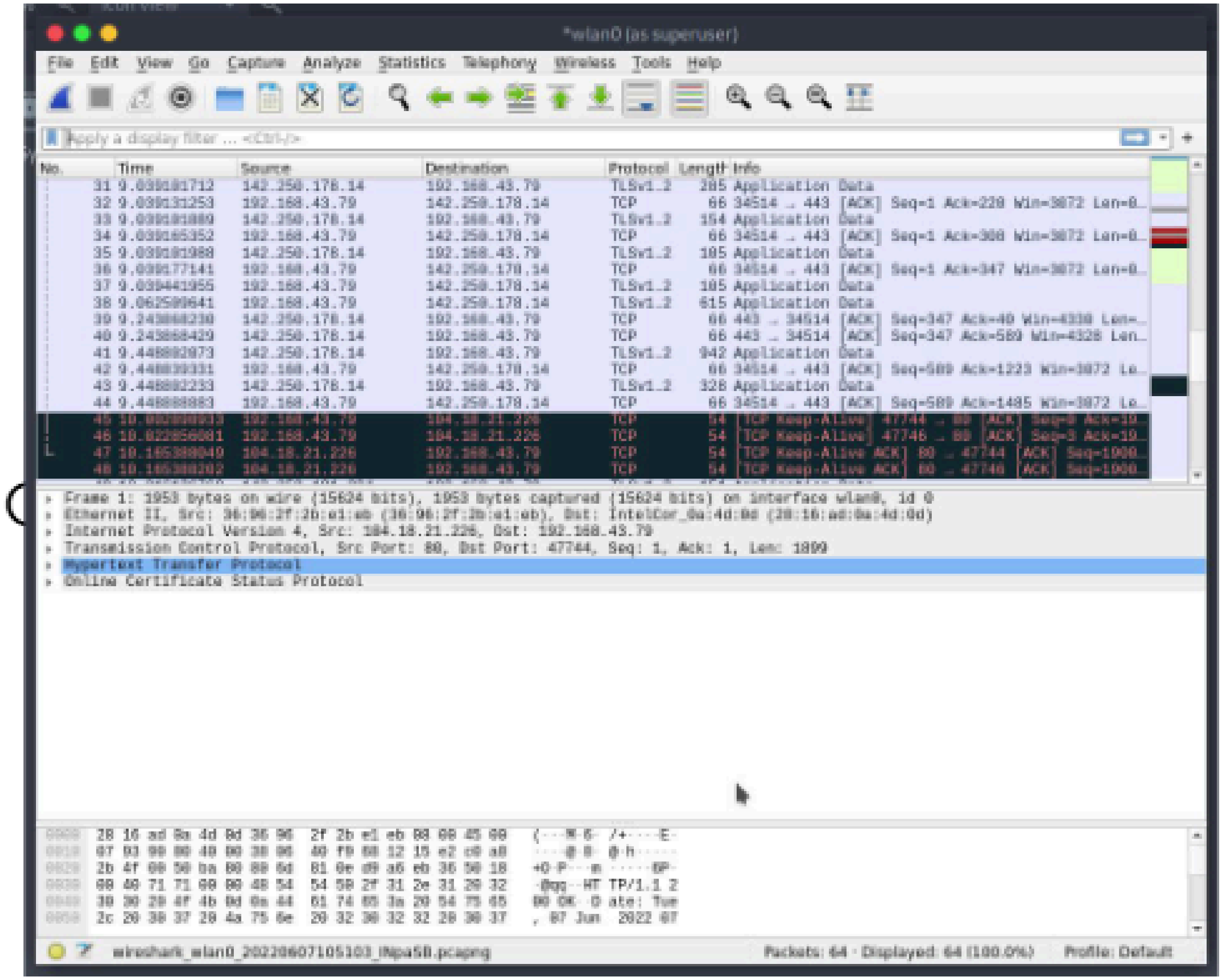
Wireshark: https://www.wireshark.org

Wireshark is a free and open-source network protocols monitoring and analysis tool.

WireShark

Core features:

- Deep inspection of hundreds of protocols.
- Live capture.
- Offline analysis.
- Support multiple OS (Windows, Linux, Mac O
- Powerful packet filters.



WireShark

Wireshark options:

- **Protocol Hierarchy**: generates a protocol hierarchy chart.

Wireshark · Protocol Hierarchy Statistics · wlan0 (as superuser)

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s
Frame	100.0	64	100.0	11196	7,088	0	0	0
Ethernet	100.0	64	8.0	896	567	0	0	0
Internet Protocol Version 4	100.0	64	11.4	1280	810	0	0	0
Transmission Control Protocol	100.0	64	80.6	9020	5,710	40	1052	666
Transport Layer Security	34.4	22	30.6	3426	2,169	22	3426	2,169
Hypertext Transfer Protocol	3.1	2	33.9	3798	2,404	0	0	0
Online Certificate Status Protocol	3.1	2	25.4	2846	1,801	2	2846	1,801

- **Conversations**: this window shows all the established conversations

Ethernet · 2 | IPv4 · 1 | IPv6 | TCP · 1 | UDP

Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
08:00:27:47:a5:3b	a4:bb:6d:e3:57:97	89	86 k	24	1,308	65	84 k	5.215093	0.0028		—
9c:65:ee:8b:47:57	ff:ff:ff:ff:ff:ff	1	60	1	60	0	0	0.000000	0.0000		—

WireShark

- **Filtering:**
  - Wireshark has powerful display filters that enables users to display what they want accurately.
- **Display Filters Operators:**

Filter	C-like	Example
equal	==	ip.src == 192.168.12.2
not equal	!=	ip.src != 192.168.12.2
AND	&&	ip.src==10.0.0.1 and tcp.flags.syn
OR	||	ip.src==10.0.0.1 or tcp.flags.syn
greater than	>	frame.len > 20
less than	<	frame.len < 100

WireShark

Protocol filters examples:

Protocol	Filters
ARP	Arp.dst.hw -> Hardware Address
HTTP	http.user_agent contains "Mozilla" -> packet user agent http.host contains "twitter" -> packet host http.request.method==GET -> request method
FTP	ftp.command -> filter FTP commands ftp.active.cip -> filter active IP address
TCP	tcp.ack -> Acknowledgement packets tcp.syn -> synchronise packets
ICMP	icmp.code -> filter ICMP code icmp.data_time -> filter ICMP data timestamp icmp.type -> filter ICMP packets types

Network Miner

- Network miner is an open source network analysis tool.

Installation: **https://www.netresec.com/?download=NetworkMiner**

Free edition features:

- Live capturing.
- PCAP parsing and analysis.
- Extract files from multiple protocol streams.
- OS Fingerprinting.

Tshark/TCPdump

tcpdump is a network packet analyzer tool that runs on command line interfaces.

- **Traffic capturing (with administrative privileges):**

tcpdump -i eth0	Capture everything.
tcpdump (src/dest) 10.0.0.2	Filter traffic by IP.
tcpdump icmp	Filter traffic by protocol.

- **PCAP analysis:**

tcpdump -n -r icmp.pcap src host 10.1.5.10	Filter by source host
tcpdump -n -r icmp.pcap net 10.10.5.0/24	Filter by source network
tcpdump -n -r icmp.pcap dst host 10.18.6.10	Filter by destination host
tcpdump -n -r icmp.pcap dst net 10.10.5.0/24	Filter by destination network